Supplementary Material

Latent-variable reanalysis of cognitive group differences in Grassi et al. (2025)

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1 Introduction

Grassi et al. (2025)'s multilab study collected the strongest dataset currently available on cognitive differences between musicians and nonmusicians. Their effort greatly advanced the older literature in sample size, coordination across laboratories, and transparency of procedures and analyses. The present reanalysis starts from that same dataset with a more focused question. Once musicians and nonmusicians differ across several cognitive measures, at what level should those differences be interpreted? General or specific abilities?

That question matters because the relevant measures are not independent of one another. Vocabulary, Raven, n-back, digit span, spatial span, and melody span all capture partly distinct task demands, but they also share variance. According to the mainstream theory on cognitive abilities, shared

variance is usually the first target of interpretation, because mean differences observed on individual tasks can reflect a broader difference in common variance rather than a series of separate task-specific effects.

Our aim, then, was not to revisit whether the original study detected group differences, it clearly did. The aim was to examine whether those differences are organized mainly at the level of a common latent factor, here labeled g, or whether they are better understood as largely independent observed-score effects.

Across both reanalyses reported below, the answer was similar. Most of the broad musician advantage was explained at the latent-factor level. At the same time, Melody span retained a clear and large residual advantage beyond g, while Wechsler Vocabulary showed a smaller residual effect in the multilevel robustness model. The resulting picture is therefore more structured than a flat task-by-task summary: a general cognitive difference, plus a limited amount of task-specific variance.

2 Method

2.1 Sample and preliminary group comparability

The reanalysis used the same matched sample analyzed in the target article, 600 musicians and 600 nonmusicians.

Before fitting latent models, we rechecked the variables that matter most for comparability across groups.

Table 1: Table S1. Preliminary group comparison for the matching variables and family SES.

Measure	Musicians	Nonmusicians	Cohen's d	95% CI
Age (years)	22.11	22.15	-0.01	[-0.13, 0.10]
Years of education	15.88	15.83	0.02	[-0.09, 0.14]
Family SES	50.06	44.90	0.34	[0.22, 0.45]

Age and years of education were essentially identical across groups, as intended by the original matching procedure. Family socioeconomic status (SES), however, was somewhat higher among musicians, with a small effect size, $d = 0.34$. For that reason, SES was retained as a covariate in the latent models.

2.2 Indicators and preprocessing

The six cognitive indicators entered in the reanalysis were Wechsler Vocabulary, Raven, n-back, Digit span, Spatial span, and Melody span. SES was included as an observed covariate.

Before the SEM analyses, the cognitive indicators and SES were standardized. This makes the model coefficients easier to read, but it also implies that not all effects live on the same scale. In particular, latent mean differences in g are expressed in latent standard deviation units because the variance of g was fixed to 1, whereas residual intercept shifts for observed variables remain in the standardized observed metric.

2.3 Correlational structure

The data show a modest but clear positive manifold. The cognitive variables are mostly positively correlated, and SES is positively related to the cognitive indicators as well.

Table 2: Table S2. Cross-group correlation matrix for the standardized cognitive indicators and family SES. The upper and lower triangles correspond to the two groups.

	Vocabulary	Raven	n-back	Digit span	Spatial span	Melody span	Family SES
Vocabulary		0.27	0.23	0.12	0.03	0.05	0.10
Raven	0.24		0.34	0.20	0.34	0.16	0.12
n-back	0.14	0.28		0.20	0.19	0.11	0.06
Digit span	0.12	0.23	0.23		0.31	0.24	0.09
Spatial span	-0.01	0.24	0.19	0.29		0.20	0.05
Melody span	0.13	0.11	0.14	0.41	0.12		0.09
Family SES	0.08	0.18	0.13	0.16	0.04	0.06	

The magnitude of the correlations is not especially large by the standards of a dedicated intelligence battery, but the overall pattern is consistent enough to motivate a common-factor representation. That point is important for interpretation: once several tasks covary positively, a task-by-task reading is no longer the only natural starting point.

2.4 Modeling rationale

Two complementary SEM strategies were used.

The first was a multigroup confirmatory factor analysis (CFA) sequence comparing musicians and nonmusicians directly. In this framework, the six cognitive scores were treated as indicators of a single latent factor, `g`, and SES was specified as a predictor of `g`. The multigroup design allowed us to examine configural, metric, and scalar invariance, and then to test whether any remaining group difference operated at the latent level.

The second was a multilevel SEM with clustering by country. This model could not reproduce the full multigroup design simultaneously, because the required multigroup-plus-multilevel specification was not feasible in `lavaan` for the present case. It provided, however, a useful robustness check that respected the clustered structure of the original dataset more directly.

2.5 How correlated residuals were handled

Because the battery had not been assembled primarily as a tightly designed psychometric battery, we did not assume that a one-factor model would absorb every residual dependency among indicators. We therefore began from a simple single-factor specification and inspected residual misfit in a transparent, limited way.

More specifically, we examined modification indices in the baseline factor model and considered adding residual covariances only when three conditions were met: the candidate covariance was among the most prominent sources of misfit, it was substantively plausible, and it improved fit without making the model needlessly opportunistic. We did not continue this process indefinitely. The goal was not to maximize fit mechanically, but to arrive at a stable and interpretable representation of the battery.

This procedure converged on the same small set of residual covariances across the main analyses: Wechsler Vocabulary with Spatial span, Spatial span with Digit span, and Digit span with Melody span. A few additional candidates appeared at isolated steps, but they were weaker or less stable and were not retained. The three selected residual covariances were then carried forward throughout the main multigroup sequence and through the multilevel comparison models. In substantive terms, these residual associations suggest that some pairs of tasks share method or content variance beyond the broad factor, which is unsurprising in a heterogeneous battery of this kind.

2.6 Model comparison logic

Model comparisons were used to guide interpretation, not to select a winner mechanically. We considered approximate fit, BIC, and interpretability jointly. When two models were numerically very close, we privileged the more transparent representation rather than over-reading negligible differences in information criteria.

3 Results

3.1 Multigroup CFA

3.1.1 Model comparison

Table 3: Table S3. Fit indices for the multigroup CFA sequence.

Model	Description	χ^2	df	p	RMSEA	CFI	BIC	ΔBIC
Configural	Same factor structure across groups	31.913	22	0.079	0.027	0.986	19525.72	53.387
Metric	Equal loadings	42.326	28	0.040	0.029	0.979	19493.60	21.259
Scalar	Equal intercepts	292.244	33	0.000	0.114	0.626	19708.07	235.728
Partial scalar	Equal intercepts except Melody span	55.698	32	0.006	0.035	0.966	19478.61	6.272
Partial scalar + equal SES -> g	Equal loadings and intercepts except Melody span, plus equal SES -> g regression	56.844	33	0.006	0.035	0.966	19472.67	0.328
Partial scalar + equal residual covariances	ScalarPartial_1 plus equal residual covariances	77.787	36	0.000	0.044	0.940	19472.34	0.000
Strict	Equal residual variances	128.276	39	0.000	0.062	0.871	19501.56	29.219
Equal latent means	Equal latent means	100.871	34	0.000	0.057	0.904	19509.60	37.264

The multigroup sequence yields a clear pattern. Configural invariance fit well, and metric invariance remained acceptable. This indicates that the

general one-factor structure is broadly similar across musicians and nonmusicians.

Full scalar invariance, by contrast, fit poorly. Once the Melody-span intercept was freed, fit improved sharply and returned to an acceptable range. Constraining the SES -> g regression to equality across groups did not materially worsen fit, and that model remained easier to interpret than the still more constrained alternative that also equated residual covariances. The latter showed a trivially smaller BIC, but the difference was negligible and approximate fit was somewhat worse. For that reason, the preferred multigroup representation was the partial-scalar model with equal SES -> g regression.

Finally, constraining the latent means to equality worsened fit again. The most direct reading is that, after accounting for measurement structure and the Melody-span intercept shift, a latent group difference remains.

3.1.2 Preferred multigroup model

Table 4: Table S4. Key estimates from the preferred multigroup model.

Parameter	Estimate	Interpretation
Latent mean difference in g (musicians - nonmusicians)	0.514	About half a latent SD
Residual mean difference in Melody span intercept	0.851	Additional Melody-span advantage beyond g, in the standardized observed metric
Standardized SES -> g (nonmusicians)	0.220	Small positive SES association with g
Standardized SES -> g (musicians)	0.198	Small positive SES association with g

The preferred multigroup model places the group difference in two locations.

First, musicians are higher on the latent factor g. The estimated latent mean difference is approximately 0.51 latent SD units, so the broad group contrast is of moderate size at the common-factor level.

Second, Melody span retains an additional intercept shift of about 0.85 units in the standardized observed metric. This means that the musician advantage is not exhausted by the common factor alone. A sizable task-specific difference remains on the indicator that is most directly related to musical material.

SES predicts g positively in both groups, and the standardized effects are very similar. Thus, the main conclusion of the multigroup analysis is not simply that musicians score higher overall, but that the broad difference is organized primarily at the latent level, with Melody span showing clear residual noninvariance beyond that general difference.

3.2 Multilevel SEM

3.2.1 Model comparison

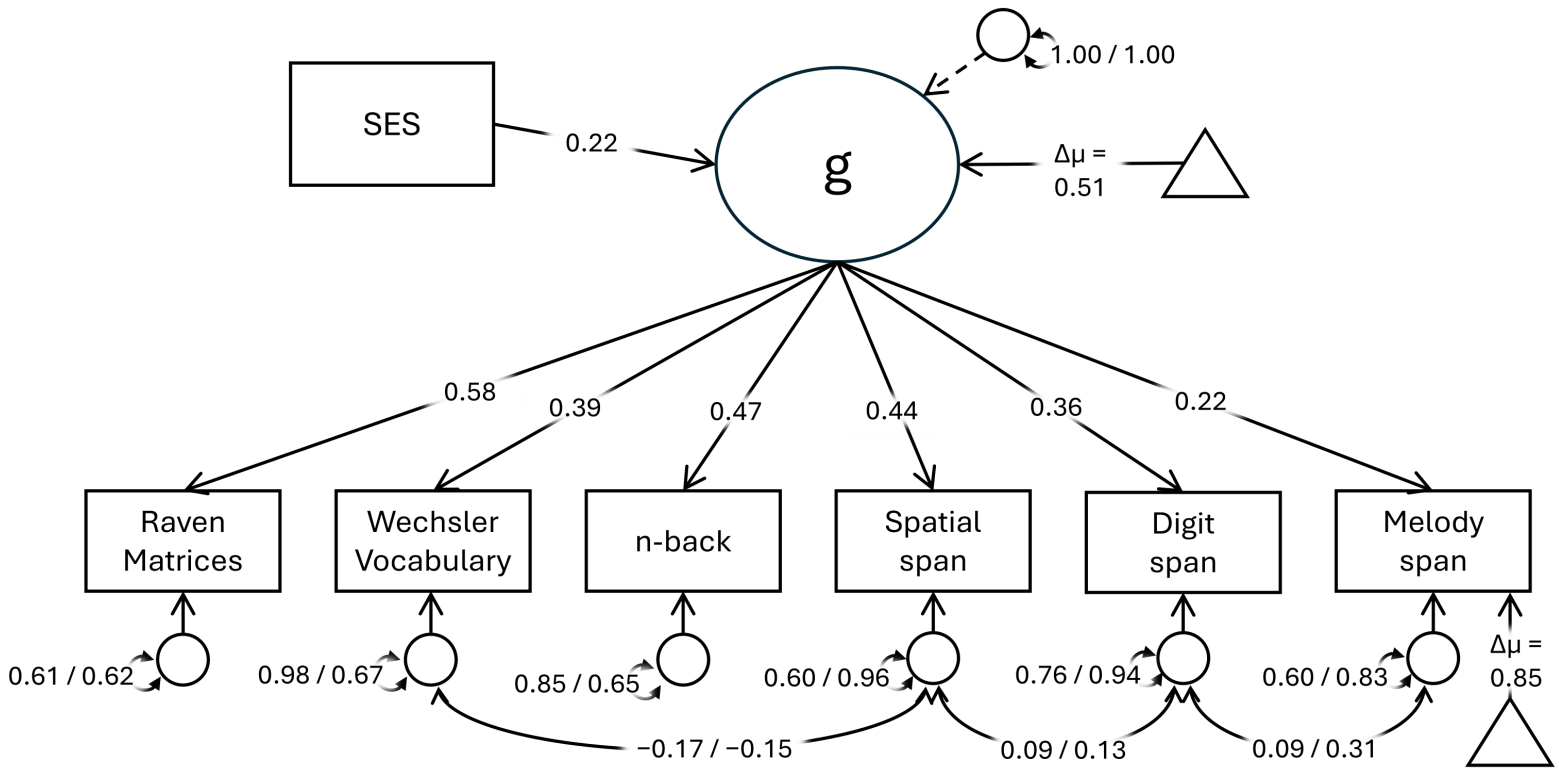

Figure 1: Figure S1. Preferred multigroup representation. Dual values separated by a slash denote the two groups for group-specific quantities such as residual variances or residual covariances. Triangles indicate where group differences in mean values emerged.

Table 5: Table S5. Fit indices for the multilevel SEM sequence, with clustering by country.

Model	Description	χ^2	df	p	RMSEA	CFI	BIC	ΔBIC
Latent g only	group differences and SES modeled only at the latent g level	297.055	31	0.000	0.085	0.777	19362.10	228.550
Observed variables only	group differences and SES modeled only at observed-variable level	34.588	21	0.031	0.023	0.989	19170.53	36.984
Latent g + broad residual paths	group differences and SES modeled at latent g plus observed-variable level (except digit span)	34.588	21	0.031	0.023	0.989	19170.53	36.984
Latent g + targeted residual paths	group differences and SES modeled at latent g plus group differences direct effects on vocabulary and melody	54.324	29	0.003	0.027	0.979	19133.55	0.000

The multilevel models were intended as a robustness check under a specification that partly respected the clustered structure of the original dataset. Here too, the ordering of models was informative.

A model placing the group effect only on `g` fit poorly. A model placing all group effects only on the observed variables fit much better. The best BIC, however, was obtained by the more selective hybrid model in which group predicted the latent factor `g` and only two residual observed paths, one to Wechsler Vocabulary and one to Melody span.

This pattern suggests that neither extreme is fully satisfactory. A pure latent-only account leaves too much systematic residual variation unexplained, whereas a pure observed-variable account is less parsimonious than needed. The targeted hybrid model sits between these extremes and provides the cleanest summary of the data.

3.2.2 Preferred multilevel model

Table 6: Table S6. Key standardized effects from the preferred multilevel model. Absolute values are reported for ease of interpretation because the sign depends on the chosen coding of the group variable.

Parameter	Estimate	Interpretation
Latent group effect on g	0.418	Latent advantage favoring musicians
Residual group effect on Wechsler Vocabulary	0.082	Small residual task-specific difference
Residual group effect on Melody span	0.430	Moderate residual task-specific difference
Standardized SES -> g	0.227	Small positive SES association with g

In the preferred multilevel model, the standardized group effect on `g` is about 0.42 SD. Beyond that, only two residual observed effects are retained: a small effect for Wechsler Vocabulary and a clearly larger one for Melody span. SES again predicts `g` positively.

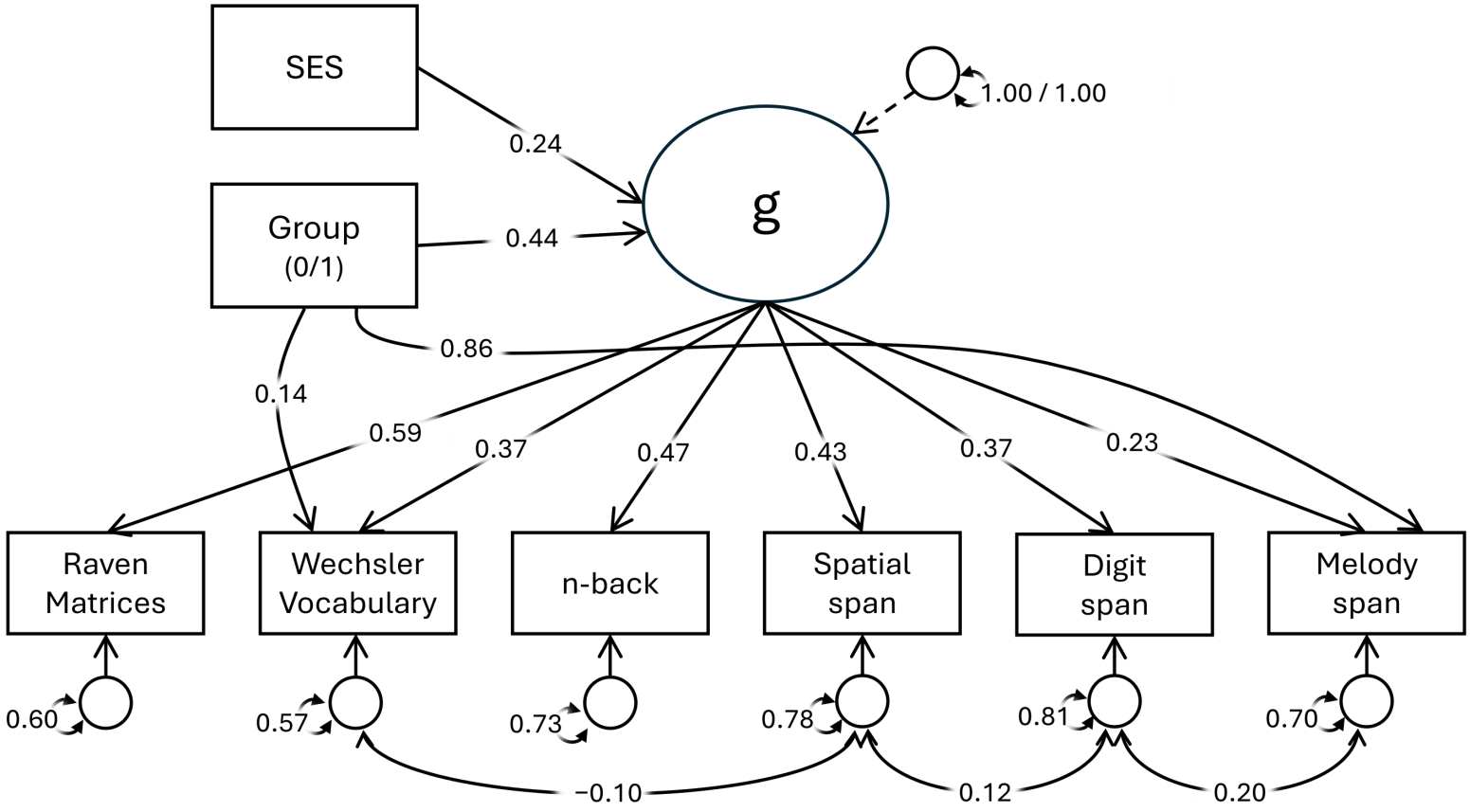

Figure 2: Figure S2. Preferred multilevel SEM representation. The sign of the raw group coefficients depends on the coding of the group variable. The diagram displays the effects in the direction favoring musicians.

The multilevel model therefore reinforces the main conclusion of the multigroup analysis while adding a small refinement. Most of the musician advantage is captured by the common factor, but not all of it. Melody span remains the clearest residual deviation, and vocabulary shows a weaker task-specific effect under the multilevel specification.

4 Discussion

Across both analytic strategies, the same broad picture emerges. The musician versus nonmusician contrast in this battery is not well summarized as a loose collection of separate observed-score differences. It is described more coherently as a difference that lies primarily at the level of shared variance among the tasks.

That conclusion matters because it changes the psychological interpretation. When the same participants show small-to-moderate advantages across several correlated tasks, the first question is not whether each task reveals its own independent substantive effect. The first question is whether the apparent multiplicity of effects mainly reflects a single broader dimension. In the present data, the answer appears to be yes.

At the same time, the latent-factor account is not exhaustive. Melody span clearly carries residual group-related variance beyond `g`, and that is theoretically unsurprising. If any indicator were expected to preserve a task-specific musician advantage after modeling common variance, it would be the measure most directly tied to musical memory. The smaller vocabulary effect that appears in the multilevel model is more tentative and should be treated as secondary to the main latent difference.

Taken together, the reanalyses therefore support a structured interpretation of the target article's findings. Musicians do not simply outperform nonmusicians on one isolated music task, nor do the data require a story in which each cognitive measure carries an equally distinct advantage. The pattern is better described as a moderate difference in broad shared cognitive variance, accompanied by a more specific residual advantage in Melody span.

4.1 Limits

The reanalysis also has limits.

First, the positive manifold is present but not especially strong, weaker than one would typically expect in a proper intelligence battery. The latent factor extracted here should therefore be read pragmatically, as a summary of shared variance in this particular set of indicators, and not necessarily coinciding with the general mental ability underlying intelligence.

Second, the battery was not originally designed around an explicit psychometric measurement model. Some residual dependencies among tasks were therefore unsurprising, and the retained correlated residuals should be understood in that light.

Third, the multilevel analysis is a robustness check rather than a full multigroup-plus-multilevel solution. It addresses clustering by country, but it cannot answer every measurement-invariance question simultaneously.

5 Reproducibility

You can find the full repository at the following URL: `https://github.com/EnricoToffalini/musicians-commentary`

The present document reads precomputed tables in `/Tables` and figures from `/Figures` subfolders from that repository. The corresponding data-preparation and full modeling, reproducible scripts are stored in the `/R` subfolder. Data was placed in the `/Data` subfolder, and correspondes to the original `dat-clean` dataset assembled by Grassi et al. (2025).